

As shown in Fig. 9, this electrode consists of a composite lithium anode with a three-layer construction. This electrode concept adopts a water-stable sodium (Na) superionic conductor (NASICON) type lithium-conducting solid glass ceramic as a protective layer that covers and isolates lithium metal from direct contact with aqueous electrolyte.

The glass ceramic is a compound of lithium (Li), titanium (Ti), aluminum (Al), and silicon (Si)—known as LTAP. But the LTAP is unstable in direct contact with lithium metal. So, a buffer layer of an electrolyte consisting of polyethelene oxide (PEO) and lithium salt is used between the lithium metal and LTAP plate. The bulk conductivity of PEO is increased by addition of nanosized ceramic fillers like BaTiO_3 .

The PEO electrolyte mixed with organic ionic liquid and nano silicon dioxide (SiO_2) particles used in the buffer layer exhibits excellent effects for the suppression of formation of lithium dendrite. Aqueous lithium-air battery of this design has a life of more than 100 cycles. For application in electric vehicles, an effective design of the battery system should discharge during acceleration or cruise and be charged by regenerative braking.

Solid-state electrolyte lithium-air battery

Amongst all types of lithium-air batteries, non-aqueous lithium-air batteries possess a relatively simple structure and have many advantages. However, despite the advantages, they exhibit ignitability, toxicity, liquid leakage, and volatilisation because of their organic liquid electrolyte components. Moreover, lithium dendrites form during cycling, resulting in internal short circuit that leads to combustion and even cell explosion.

More seriously, the charging process is accompanied by the decomposition of organic electrolytes, leading to the accumulation of side prod-

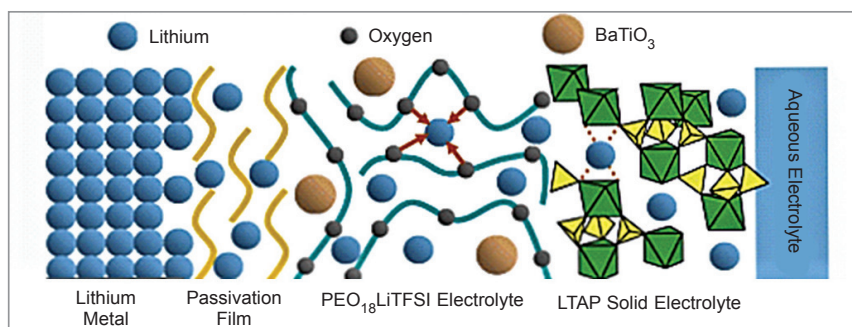


Fig. 9: Composite lithium anode with three-layer construction (Credit: The Chemical Society Of Japan, 2011)

ucts, gradually increasing overpotential and finally death of the cell. When operated in ambient air, corrosion of lithium metal occurs because water and other gases pass through the open structure of the air electrode and organic electrolyte.

These problems result in poor cycle and rate performance. One of the optimal methods to offset these disadvantages is developing solid-state lithium-air batteries, replacing the organic electrolytes with solid electrolytes, such as inorganic or polymer electrolytes. Fig. 10, shows a typical schematic arrangement of solid-state electrolyte lithium-air battery consisting of a lithium metal anode, a solid-state electrolyte, and a porous carbon cathode with suitable ionic and electronic paths, which have the potential to eliminate the aforesaid problems.

There is another advantage that these batteries don't require separator. They can work over wide operating temperatures. The main drawback of this system is low conductivity of the ceramic electrolyte compared to aqueous and aprotic batteries.

Solid-state electrolyte is a solid ionic conductor capable of conducting ions in the solid state. It should have ionic conductivity, at least higher than $10^{-4} \text{ S cm}^{-1}$, high ionic transference number (closest

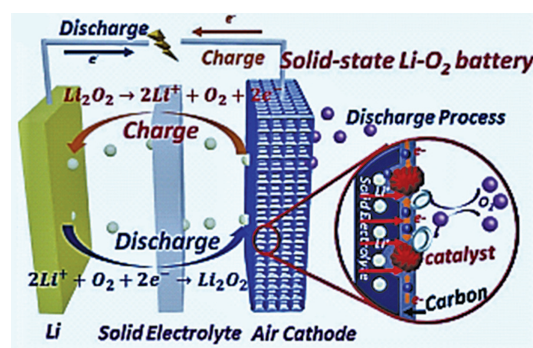


Fig. 10: Schematic arrangement of solid-state lithium-air battery

possible to 1), wide electrochemical stability window (ESW), at least 4 to 5V, and high mechanical strength of about tens of MPa.

Solid-state electrolytes can be divided into following categories:

1. Inorganic solid electrolyte (ISE) that is constituted by an inorganic material in the crystalline or glassy state, which conducts ions by diffusion through the lattice. These have high ionic conductivity at room temperature, high modulus of the order of GPa, and high transfer number. But they are generally brittle. They can be LISICON (lithium superionic conductor, for example, LGPS and LiSiPS) or NASICON (sodium superionic conductor like LAGP and LATP).

NASICON type solid electrolyte has a 3D structure with a tunnel for sodium ion transport. In the lithium ionic conductors, sodium ions are substituted with lithium ions in a similar structure. Among these compounds, $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (LATP) ceramic reached a high ionic